

ANSH THAPA

B.TECH

CU24250224

Experiment No. 2: Contrast Enhancement and Histogram-Based Image Processing using Python and OpenCV

Aim: To analyze and implement contrast enhancement techniques using histogram analysis and histogram equalization in Python and OpenCV for improving image quality and enhancing visual information in low-contrast images.

Purpose

Image contrast plays a significant role in the performance of computer vision systems. Poor illumination, sensor limitations, or environmental conditions often result in low-contrast images, making feature extraction and object recognition difficult. Histogram-based enhancement techniques improve the distribution of pixel intensities, thereby enhancing image visibility and facilitating subsequent image analysis tasks.

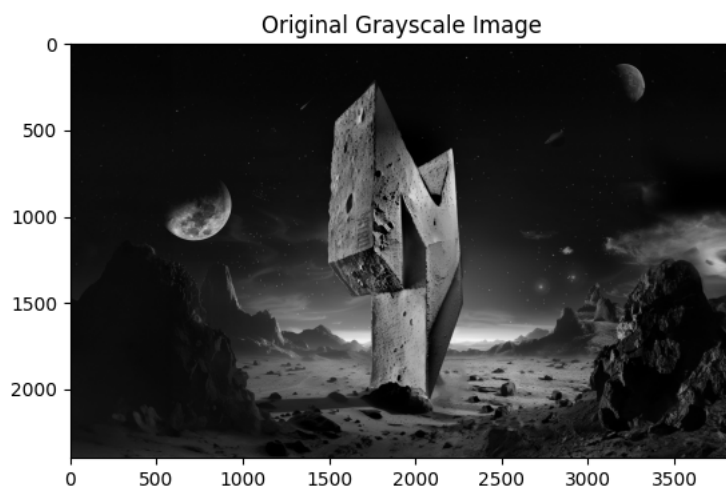
In this experiment, students will implement contrast stretching, histogram analysis, and histogram equalization using Python and OpenCV. The effectiveness of these enhancement techniques will be evaluated on real-world images such as medical images, satellite images, or low-light photographs.

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
img = cv2.imread('photo.jpg')
print(img is None)
```

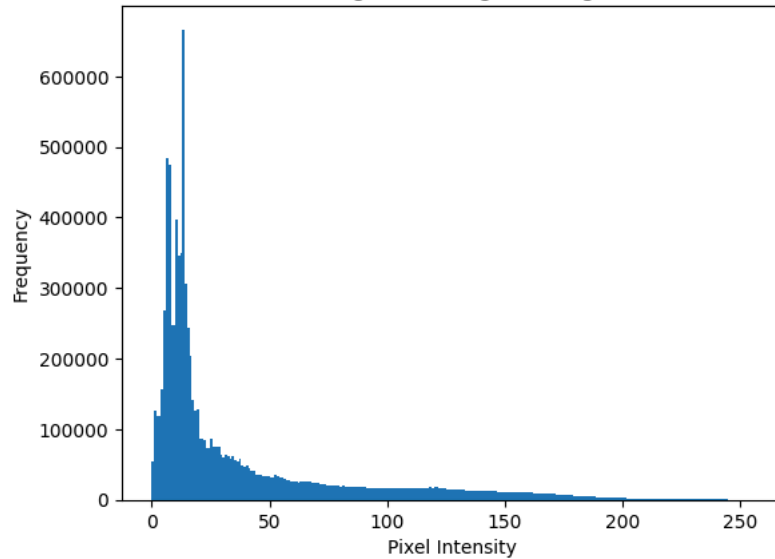
```
False
```

```
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
plt.imshow(gray, cmap='gray')
plt.title('Original Grayscale Image')
plt.show()
```



```
plt.hist(gray.ravel(), bins=256, range=[0,256])
plt.title('Histogram of Original Image')
plt.xlabel('Pixel Intensity')
plt.ylabel('Frequency')
plt.show()
```

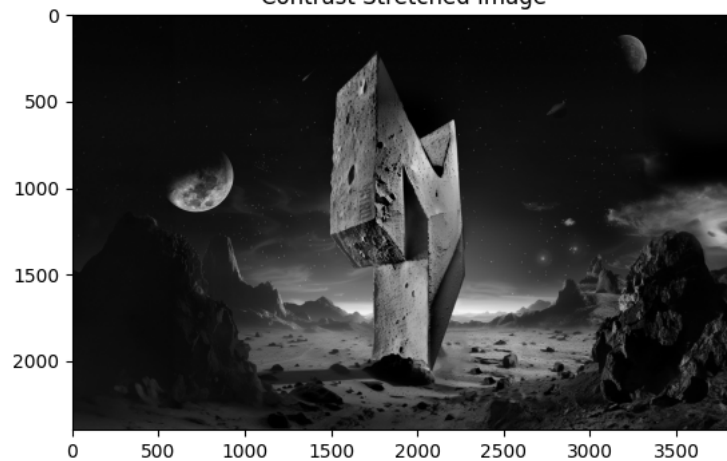
Histogram of Original Image



```
min_val = np.min(gray)
max_val = np.max(gray)
stretched = ((gray - min_val) / (max_val - min_val) * 255).astype(np.uint8)

plt.imshow(stretched, cmap='gray')
plt.title('Contrast Stretched Image')
plt.show()
```

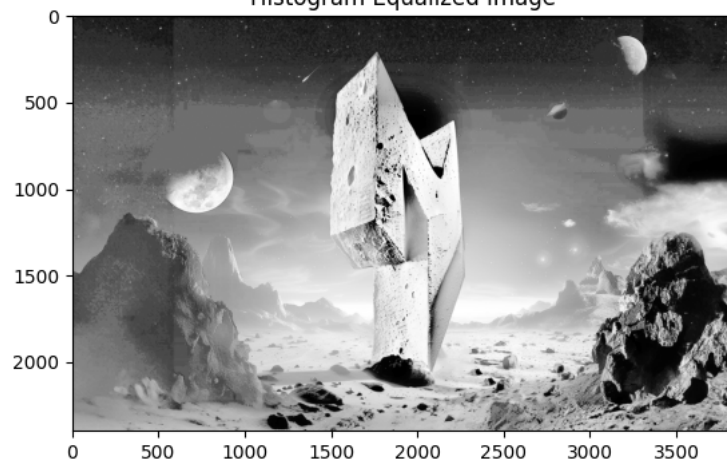
Contrast Stretched Image



```
equalized = cv2.equalizeHist(gray)

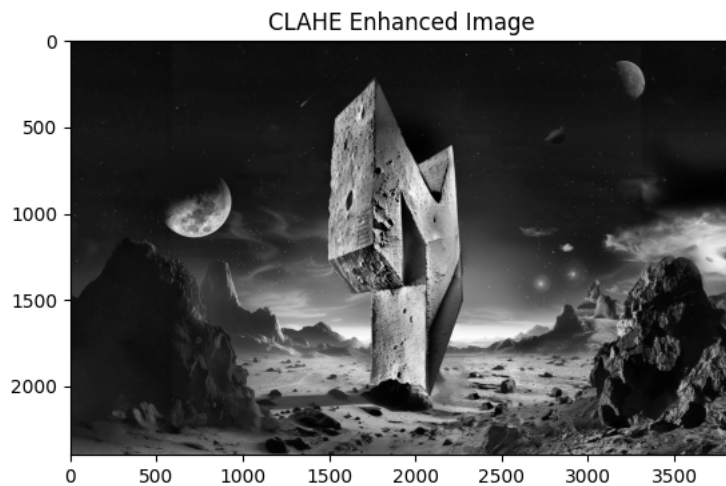
plt.imshow(equalized, cmap='gray')
plt.title('Histogram Equalized Image')
plt.show()
```

Histogram Equalized Image



```
clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8,8))
clahe_img = clahe.apply(gray)

plt.imshow(clahe_img, cmap='gray')
plt.title('CLAHE Enhanced Image')
plt.show()
```



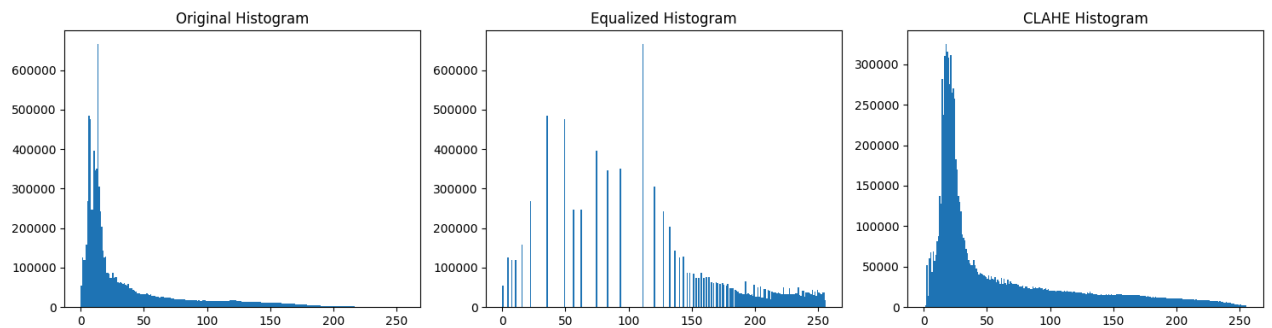
```
fig, axs = plt.subplots(1, 3, figsize=(15,4))

axs[0].hist(gray.ravel(), bins=256, range=[0,256])
axs[0].set_title('Original Histogram')

axs[1].hist(equalized.ravel(), bins=256, range=[0,256])
axs[1].set_title('Equalized Histogram')

axs[2].hist(clahe_img.ravel(), bins=256, range=[0,256])
axs[2].set_title('CLAHE Histogram')

plt.tight_layout()
plt.show()
```



```
fig, axs = plt.subplots(2, 2, figsize=(10, 8))
titles = ['1. Original', '2. Contrast Stretched', '3. Histogram Equalized', '4. CLAHE']
images = [gray, stretched, equalized, clahe_img]

for ax, img_, title in zip(axs.flat, images, titles):
    ax.imshow(img_, cmap='gray')
    ax.set_title(title)
    ax.axis('off')

plt.subplots_adjust(hspace=0.1, wspace=0.1)
plt.show()
```

1. Original



2. Contrast Stretched



3. Histogram Equalized



4. CLAHE



Questions

1. What is image contrast, and why is it important in image processing?

Image contrast is the difference between the brightest and darkest pixel values in an image. It is important because a well-contrasted image allows objects, edges, and features to stand out clearly, making it easier for both the human eye and computer vision algorithms to detect and analyze them. Low-contrast images have details that blend together, making feature extraction difficult.

2. Explain the concept of an image histogram. What information does it provide?

An image histogram is a graph that shows how many times each pixel intensity value (0 to 255 in grayscale) occurs in an image. The x-axis represents intensity values and the y-axis represents their frequency. It provides information about the overall brightness, contrast, and pixel distribution of an image — for example, a histogram concentrated on the left indicates a dark image, one concentrated on the right indicates a bright image, and one spread across the full range indicates good contrast.

3. Differentiate between Histogram Stretching and Histogram Equalization.

Histogram Stretching (Contrast Stretching) is a linear technique that stretches the existing pixel range (min-max) across the full 0-255 range without changing the shape of the distribution. Histogram Equalization is a non-linear technique that uses the cumulative distribution function (CDF) to redistribute pixel intensities so the histogram becomes as flat/uniform as possible — this increases contrast more aggressively but can sometimes look unnatural.

4. What is Contrast Limited Adaptive Histogram Equalization (CLAHE)? How does it differ from standard Histogram Equalization?

CLAHE is an improved version of standard Histogram Equalization that divides the image into small tiles (blocks) and applies equalization separately to each tile, rather than to the whole image at once — this handles local contrast better. It also uses a "clip limit" that prevents noise from being over-amplified. Standard HE is a global operation (applied uniformly to the whole image), while CLAHE is local/adaptive (applied differently per region), so CLAHE generally produces more natural and balanced results, especially for images with uneven lighting.

5. Why is histogram equalization commonly applied before feature extraction and image segmentation?

Histogram equalization improves image contrast, making edges, textures, and boundaries more prominent. This helps feature extraction algorithms (such as edge detection) and segmentation techniques perform better because objects become more clearly distinguishable from the background, improving overall accuracy.

6. Mention three real-world applications where histogram equalization is widely used.

- Medical imaging (X-rays, MRI, CT scans) to clearly reveal tissues/abnormalities
- Satellite and remote sensing images for better terrain feature analysis
- Low-light photography and surveillance cameras (night vision, CCTV footage enhancement)

7. What are the limitations of global histogram equalization?

- It applies a single transformation to the whole image, so local details in different regions are ignored
- It can amplify noise, especially in uniform (flat) regions
- The image can become over-enhanced, resulting in an unnatural or "washed out" look
- It performs poorly on images with uneven lighting (parts of the image bright, other parts dark)

8. How does contrast enhancement improve the performance of object detection and recognition systems?

Contrast enhancement makes the differentiation between objects and their background clearer, sharpening edges and boundaries. This helps detection algorithms (such as CNN-based models) extract features more effectively, reduces false negatives, and improves accuracy even under poor lighting conditions.

9. Compare histogram-based enhancement techniques with brightness adjustment methods.

Brightness adjustment simply adds or subtracts a constant value to/from all pixels — this makes the whole image uniformly brighter or darker, but the contrast (the difference between dark and light areas) remains the same. Histogram-based techniques (like HE, CLAHE) dynamically redistribute the pixel distribution, actually improving contrast rather than just overall brightness — making them more effective for improving image quality.

10. Why is CLAHE preferred for medical imaging and low-light image enhancement?

Medical images (X-ray, MRI) contain subtle details and local variations (such as tumors, tissue boundaries) that are important and could be over-enhanced or lost with global HE. CLAHE preserves these fine details by operating on local tiles without excessively amplifying noise, and the clip limit prevents over-enhancement. Similarly, in low-light images, different regions have different lighting conditions, which CLAHE handles better than standard HE.